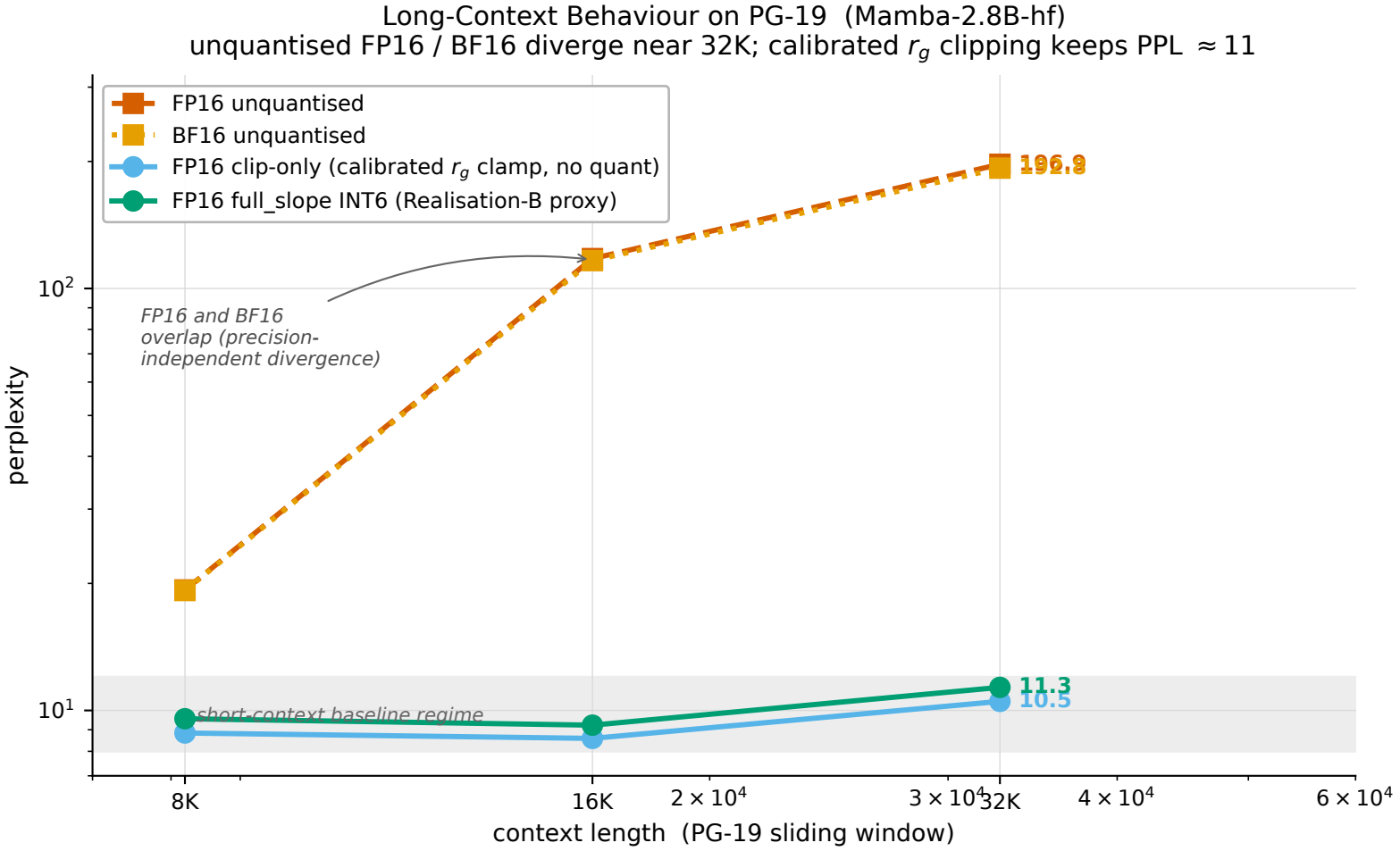
Long-Context Behaviour on PG-19 (Mamba-2.8B-hf)
unquantised FP16 / BF16 diverge near 32K; calibrated r_g clipping keeps PPL ≈ 11
FP16 unquantised
BF16 unquantised
FP16 clip-only (calibrated r_g clamp, no quant)
FP16 full_slope INT6 (Realisation-B proxy)
196.9
192.8
FP16 and BF16
overlap (precision-
independent divergence)
11.3
10.5
short-context baseline regime
10^2
10^1
perplexity
8K
16K
2×10^4
3×10^4
32K
4×10^4
6×10^4
context length (PG-19 sliding window)